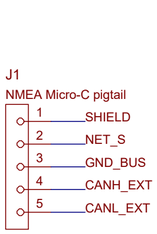
ORION BRIDGE R4 / NMEA 2000 - BLE

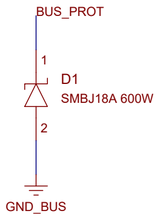
23.09.2026 / Altium / JLCPCB

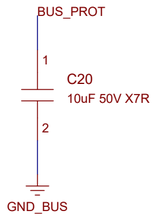
01 Versorgung / 9-16 V NMEA 2000 / isolierte Logik

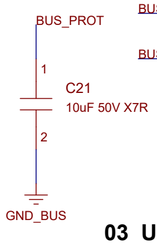
01 Anschluss, Verpolschutz und geregelte 5-V-Busversorgung

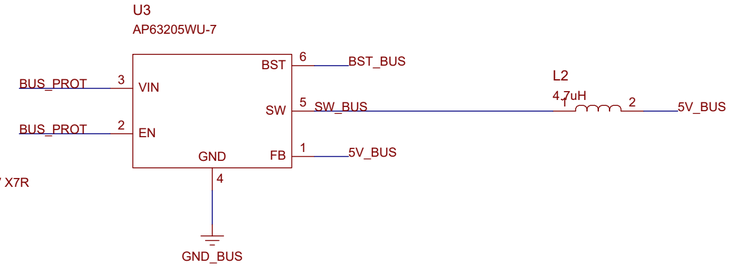


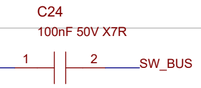


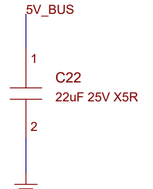


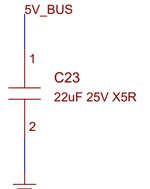


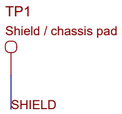




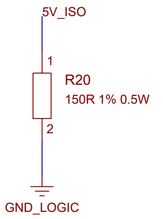


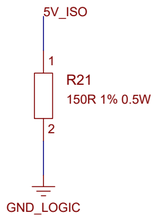




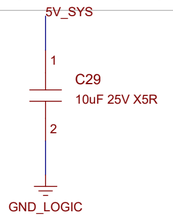


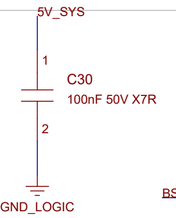
02 Galvanische Trennung und Mindestlast

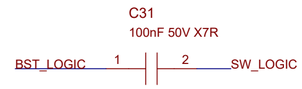


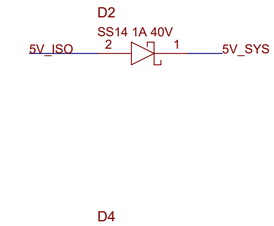


03 USB-Entkopplung und 3,3-V-Logikversorgung

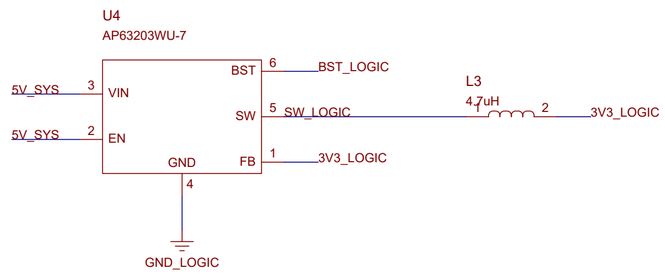


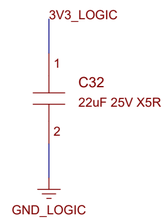


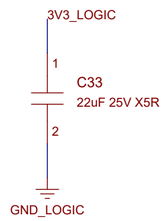












NET-S: 9-16 V DC gegen NET-C (GND_BUS). Konservatives Firmware-Buslastbudget: LEN 12.

GND_BUS und GND_LOGIC bleiben galvanisch getrennt. USB-GND und USB-Schirm gehoeren zu GND_LOGIC.

D2/D4 entkoppeln die Versorgung automatisch. R20/R21 stellen die Mindestlast des isolierten Wandlers bereit.

J1: Kabel-Loetanschluss; Pin 1 Schirm, 2 NET-S, 3 NET-C, 4 CAN-H, 5 CAN-L. Keine 120-Ohm-Terminierung.

Geometrievorschau aus SchDoc-Daten | Kein Altium-Rendering und keine ERC-/DRC-Pruefung